

Fig. 10.2 Narrow gap-welded crankshaft fabricated with cast steel sections.

Another requirement of a welded crankshaft is for the journals and pins to be hollow. Backing rings are then used to obtain alignment and for the start of the weld. When welding is finished the ring and the initial passes are machined out to remove any doubtful material from the inner part of the weld.

The usual heat treatment and testing required for high quality welding are carried out prior to finish-machining.

The main advantages of welded crankshafts are savings in weight when compared with crankshafts constructed in the normal manner by shrink-fit assembly, lower cost arising out of the reduced size of parts being forged, and ease of forging the pieces to obtain good grain-flow characteristics in the finished components.

The sectional profile of a welded crankshaft can be made similarly to that of a 'cast-iron' crankshaft. That is, the main masses of metal are located in those regions where the highest stresses would normally occur. (Fig. 10.2)

10.8 What state of stress exists in a web and crankpin or journal of a built-up crankshaft?

The state of the stresses can be visualized if we consider the sizes of these parts before they are heated and shrunk together and then relate these sizes to those found when shrinking has taken place and the parts have cooled to normal temperature.

After the shrink fit of the pin or journal is made in a crank web the diameter of the pin or journal will be reduced in the way of the shrink; similarly the diameter of the hole in the web will be increased.

During the cooling process the heated web will decrease in size and start compressing the surface of the journal; this will continue until a balance point is reached, when the force per unit area between the journal surface and the surface of the hole in the web will be the same and will be acting radially. Thus there will be a radial compressive stress acting on the mating surfaces of the pin and the hole.

If we were to take a cast-iron washer and drive a steel taper pin in the hole, the washer would eventually split along a line radial in direction and then open up, failure of the washer being due to tensile loading. In driving the taper pin into the washer we are simulating the conditions existing in the shrink fit, so in a

similar manner we create a hoop tensile stress in the web after it has cooled down. As the radial thickness of the web around the hole is large, the radial compressive stress and the hoop tensile stress are more concentrated at the surface of the hole and they both diminish gradually in an outward direction.

Note For a mathematical treatise on the stresses in crank webs due to shrink fits, see any textbook on the strength of materials which deals with the theorem of Lamé, and pressures in thick cylinders and compound tubes.

10.9 What would be the result if the shrinkage allowance of crankshaft crankpins and journals were (a) too great, and (b) too small?

If the shrinkage allowance is too great there is a danger of the crankpins and journals becoming slack in the webs, while if the allowance is too small the material of the webs around the holes may be overstressed and the elastic limit of the material reached. Steel in such a condition will quickly fracture.

Note The elastic limit of steel or similar materials is reached when the relationship between stress and strain ceases to follow Hooke's Law. Beyond the elastic limit steel begins to yield under the action of the load (see also Questions 6.44, 6.45 and 10.8).

10.10 What is the usual shrinkage allowance for the shrink fits of built-up crankshafts?

The shrinkage allowance for both the pins and journals of built-up crankshafts is the same and is between 1/570 and 1/660 of the diameter of the pin or journal.

Example. Find the bore diameter to which a web must be machined to accommodate a crank pin or journal of 400 mm diameter.

The size of the hole (lower limit) is

$$400 - 400/570 = 400 - 0.70 = 399.30 \text{ mm}$$

and the size of the hole (upper limit) is

$$400 - 400/660 = 400 - 0.61 = 399.39 \text{ mm}$$

The mean bore size is therefore

$$\frac{399.39 + 399.30}{2} = 399.345 \text{ mm}$$

and the bore-size hole in the web is

$$399.35(\pm 0.02) \text{ mm}$$

10.11 What are the proportions of crank webs relative to the pin or journal diameter in built crankshafts?

The proportions of crank webs must be such that the resultant shrink fit enough grip without overstressing the material in the web.